

Revised Generation Interconnection System Impact Study Report

For Queue Project Q22-2089

Gisler 138 kV
May 02, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2089. The Interconnection Customer proposes a 177.3 MW battery energy storage facility interconnecting at the Gisler 138 kV substation in the Pennsylvania area. The request was received on May 20, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 495 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$90,471,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Heardmont - Urbana Perkins Rd 138kV	White Swan - Gisler 138kV	128.3	165	100.0
Heardmont - Nueva Guinea 3405 138kV	White Swan Substation - Gisler 138kV	125.7	175	100.0
Clear Springs 345/138 kV XFMR	W Gallatin - Wlnut 345kV	124.2	480	6.5
Nueva Guinea 3405 - Westside 138kV L	White Swan - Gisler 138kV Line	115.9	196	100.0
Verde Mar 2605 - Croton Falls 138kV L	W Gallatin Substation - Walnut Substation 345kV	114.1	512	7.4
Flagstaff - Alliant Sbd 9801 Aegon Martha'S Way 138kV	Wallenpaupack Substation 500-345 kV XFMR	113.9	376	7.1
Split Rock - Judith Gap Wind Farm 138kV	Walnut - W Gallatin 345kV	112.8	316	5.1
Pearl - Sorrell Hill 138kV L	Pekville - Pearl Substation 138kV	110.3	287	6.4
Pearl - Alliant Sbd 9801 Aegon Martha'S Way 138kV	Wallenpaupack 500/345kV Auto	108.8	405	7.1
Verde Mar 2605 - Juith Gap Wind Farm 138kV	Walnut - West Gallatin 345kV	106.7	340	5.1
White Swan 345/138kV Auto	W Gallatin - Walnut 345kV Line	106.7	422	39.2
Railhead - S Mountain 138kV	Kookia Substation - Esquon 138kV	106.4	312	12.7
Urbana Perkins Rd - Gisler 138kV Line	White Swan - Gisler 138kV	106.3	192	100.0
Westside - White Bluffs 138kV	White Swan - Gisler 138kV Line	105.9	224	100.0
Pearl - Peckville 138kV	Pearl - Sorrell Hill 138kV	105.4	327	6.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Blackbrook Sw Sta - Hale 138kV L	Alliant Sbd 9801 Aegon Martha'S Way - Pearl 138kV Line	105.4	173	7.7
Breed - Sidney A Murray Jr Hydroelectric 230kV	Richland - Dekalb 500kV	101.8	2,568	5.5
Talenenergy Martins Creek Llc Harwood 500 kV/1 kV Autotransformer	Felida 500/345 kV Transformer	100.7	1,746	13.8
Talenenergy Martins Crk Llc Harwood 345/1kV TX	Felida 500/345 kV Autotransformer	100.7	1,746	13.8
Wheelabrator Frackville Energy - Upsco 345kV Line	Morganville - Westwood Generation Llc 345kV L	100.4	1,610	9.1
White Swan - Gisler 138kV	Polk St - Railhead 138kV line	100.4	239	74.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Heardmont - Urbana Perkins Rd 138kV	Rebuild line with higher-capacity conductor	\$5,024,000	100% (\$5,024,000)
Nueva Guinea 3405 - Heardmont 138kV line	Rebuild line with higher-capacity conductor	\$3,990,000	100% (\$3,990,000)
Clear Springs 345 kV/138 kV Transformer	Replace transformer with higher-rated unit	\$20,783,000	42% (\$8,814,000)
Nueva Guinea 3405 - Westside 138kV L	Rebuild line with higher-capacity conductor	\$2,161,000	100% (\$2,161,000)
Verde Mar 2605 - Croton Falls 138kV L	Rebuild line with higher-capacity conductor	\$7,478,000	100% (\$7,478,000)
Flagstaff - Alliant Sbd 9801 Aegon Martha'S Way 138kV	Rebuild line with higher-capacity conductor	\$2,242,000	100% (\$2,242,000)
Judith Gap Wind Farm - Split Rock 138kV L	Rebuild line with higher-capacity conductor	\$10,713,000	100% (\$10,713,000)
Pearl - Sorrell Hill 138kV L	Rebuild line with higher-capacity conductor	\$2,806,000	100% (\$2,806,000)
Pearl - Alliant Sbd 9801 Aegon Martha'S Way 138kV	Reconductor line	\$839,000	100% (\$839,000)
Verde Mar 2605 - Juith Gap Wind Farm 138kV	Reconductor line	\$5,512,000	100% (\$5,512,000)
White Swan Substation 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$17,563,000	100% (\$17,563,000)
Railhead - S Mountain 138kV	Reconductor line	\$1,398,000	74% (\$1,030,000)
Urbana Perkins Rd - Gisler 138kV Line	Reconductor line	\$3,937,000	100% (\$3,937,000)
Westside - White Bluffs 138kV	Reconductor line	\$7,231,000	100% (\$7,231,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pearl - Peckville 138kV	Reconductor line	\$1,036,000	100% (\$1,036,000)
Blackbrook Sw Sta - Hale 138kV L	Reconductor line	\$7,034,000	100% (\$7,034,000)
Breed - Sidney A Murray Jr Hydroelectric 230kV	Reconductor line	\$16,427,000	2% (\$276,000)
Talenenergy Martins Crk Llc Harwood 500 kV/1 kV Auto T1	Replace transformer with higher-rated unit	\$27,033,000	4% (\$1,036,000)
Talenenergy Martins Crk Llc Harwood 345/1kV TX	Replace transformer with higher-rated unit	\$14,209,000	4% (\$544,000)
Upsco - Wheelabrator Frackville Energy 345kV	Reconductor line	\$8,023,000	4% (\$279,000)
White Swan - Gisler 138kV	Reconductor line	\$1,045,000	89% (\$926,000)
	Total allocated to project		\$90,471,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.